

33.3 Repeated measures

A repeated-measures study observes the same subject on several occasions: a clustered design whose cluster is the subject and whose “treatments” are the occasions, laid out like the split-plot design of [Section 33.1](#), and for many years analysed that way.

The analogy has one weak joint. What makes the split-plot covariance compound symmetric is the *randomization* of subplot treatments within the whole plot, and occasions cannot be randomized: measurements close together in time are more alike than distant ones, so compound symmetry here is usually false. What is remarkable is that the univariate F test survives a much weaker condition, and that the condition can be written down exactly.

33.3.1 The univariate analysis

Let $\mathbf{y}_1, \dots, \mathbf{y}_N$ be independent $N_m(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ vectors, one per subject, with $\boldsymbol{\Sigma}$ positive definite and unknown, and write y_{ij} for occasion j of subject i . The hypothesis of interest is

$$H : \mu_1 = \dots = \mu_m, \quad \text{that is,} \quad \boldsymbol{\mu} \in \text{span}\{\mathbf{1}\}.$$

Laid out with subjects as rows and occasions as columns the data have one observation per cell, and the classical analysis is the additive two-way analysis of variance of [Section 16.1](#), with the subject-by-occasion interaction as error. With $\bar{y}_{i\cdot}$, $\bar{y}_{\cdot j}$ and $\bar{y}$ the subject, occasion and grand means,

$$SS_T = N \sum_{j=1}^m (\bar{y}_{\cdot j} - \bar{y})^2, \quad SS_E = \sum_{i=1}^N \sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y})^2, \quad (33.3.1)$$

and the statistic is $F = \{SS_T/(m-1)\}/\{SS_E/((N-1)(m-1))\}$. It is the subplot stratum of a split-plot design with one whole plot per subject, and under compound symmetry it is exact ([Theorem 33.3.2\(b\)](#) below, with [Exercise \(B1\)](#)). The question is how much of that exactness survives a general $\boldsymbol{\Sigma}$.

Everything reduces to the within-subject contrasts. Let $\mathbf{U}$ be $m \times (m-1)$ with orthonormal columns orthogonal to $\mathbf{1}$, so that $\mathbf{U}\mathbf{U}^\top = \mathbf{I} - \bar{\mathbf{J}}_m$, and set $\mathbf{z}_i = \mathbf{U}^\top \mathbf{y}_i$: independent $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \boldsymbol{\Psi})$ with $\boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}$, with H saying exactly that $\mathbf{U}^\top \boldsymbol{\mu} = \mathbf{0}$. The subject means carry no information about H .

33.3.2 Sphericity

Definition 33.3.1. Sphericity and Box's measure

Let $\boldsymbol{\Sigma}$ be an $m \times m$ positive definite matrix and $\mathbf{U}$ as above. $\boldsymbol{\Sigma}$ satisfies **sphericity** if $\mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U} = \lambda \mathbf{I}_{m-1}$ for some $\lambda > 0$. **Box's measure of sphericity** is

$$\epsilon(\boldsymbol{\Sigma}) = \frac{\{\text{tr } \boldsymbol{\Psi}\}^2}{(m-1) \text{tr}(\boldsymbol{\Psi}^2)}, \quad \boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}. \quad (33.3.2)$$

Neither the condition nor ϵ depends on the choice of $\mathbf{U}$: two such matrices satisfy $\tilde{\mathbf{U}} = \mathbf{U}\mathbf{Q}$

with $\mathbf{Q}$ orthogonal, so $\tilde{\Psi} = \mathbf{Q}^\top \Psi \mathbf{Q}$ has the same trace, the same trace of square, and is a multiple of the identity exactly when Ψ is.

Lemma 33.3.1. Equivalent forms of sphericity

For a positive definite $m \times m$ matrix Σ the following are equivalent:

- (i) $\mathbf{U}^\top \Sigma \mathbf{U} = \lambda \mathbf{I}_{m-1}$;
- (ii) $\text{Var}(Y_j - Y_k) = 2\lambda$ for every $j \neq k$;
- (iii) $\sigma_{jk} = a_j + a_k + \lambda \delta_{jk}$ for some $a_1, \dots, a_m$, where δ_{jk} is one if $j = k$ and zero otherwise;
- (iv) $\Sigma = \lambda \mathbf{I} + \mathbf{1}\mathbf{a}^\top + \mathbf{a}\mathbf{1}^\top$ for some $\mathbf{a} \in \mathbb{R}^m$.

Moreover $1/(m-1) \leq \epsilon(\Sigma) \leq 1$, with $\epsilon = 1$ iff sphericity holds, and compound symmetry implies sphericity but not conversely.

Proof. Write $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}_m = \mathbf{U}\mathbf{U}^\top$. Since $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_{m-1}$, condition (i) holds iff $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$.

- (i) $\Rightarrow$ (ii): $\mathbf{e}_j - \mathbf{e}_k$ is orthogonal to $\mathbf{1}$, hence fixed by $\mathbf{C}$, so $\text{Var}(Y_j - Y_k)$ equals $(\mathbf{e}_j - \mathbf{e}_k)^\top \mathbf{C}\Sigma\mathbf{C}(\mathbf{e}_j - \mathbf{e}_k)$, which is $\lambda\|\mathbf{e}_j - \mathbf{e}_k\|^2 = 2\lambda$.
- (ii) $\Rightarrow$ (iii): $\text{Var}(Y_j - Y_k) = \sigma_{jj} + \sigma_{kk} - 2\sigma_{jk} = 2\lambda$ gives $\sigma_{jk} = (\sigma_{jj} - \lambda)/2 + (\sigma_{kk} - \lambda)/2$ for $j \neq k$. Put $a_j = (\sigma_{jj} - \lambda)/2$; then $\sigma_{jk} = a_j + a_k$ off the diagonal and $\sigma_{jj} = 2a_j + \lambda$ on it, which is (iii).
- (iii) $\Leftrightarrow$ (iv) is the same statement in matrix form.
- (iv) $\Rightarrow$ (i): $\mathbf{C}\mathbf{1} = \mathbf{0}$, so $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$.

For the bounds, let $\psi_1, \dots, \psi_{m-1} > 0$ be the eigenvalues of Ψ . Then $\epsilon = (\sum_l \psi_l)^2 / \{(m-1) \sum_l \psi_l^2\}$. The Cauchy–Schwarz inequality gives $(\sum_l \psi_l)^2 \leq (m-1) \sum_l \psi_l^2$ with equality iff all ψ_l are equal, so $\epsilon \leq 1$ with equality iff $\Psi = \lambda \mathbf{I}$; and $\sum_l \psi_l^2 \leq (\sum_l \psi_l)^2$ for nonnegative numbers gives $\epsilon \geq 1/(m-1)$. Compound symmetry is (iv) with all a_j equal; taking the a_j unequal gives sphericity without compound symmetry, and such a Σ is positive definite as soon as λ is large enough. ■

Condition (ii) is the form to remember and to check: *all pairwise differences of the repeated measurements have the same variance*. Compound symmetry says more than the F test needs.

Theorem 33.3.2. Sphericity is necessary and sufficient for the exact F test

Let $\mathbf{y}_1, \dots, \mathbf{y}_N$ be independent $N_m(\boldsymbol{\mu}, \Sigma)$ with Σ positive definite, $m \geq 2$, $N \geq 2$, and let F be the statistic built from Eq. 33.3.1. Put $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$, $\mathbf{q} = \sqrt{N} \bar{\mathbf{z}}$ and $\mathbf{W} = \sum_i (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})^\top$.

- (a) $\text{SS}_T = \|\mathbf{q}\|^2$ and $\text{SS}_E = \text{tr}(\mathbf{W})$, and under H the vector $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$ is independent of $\mathbf{W} \sim W_{m-1}(N-1, \Psi)$. Hence $F = N(N-1)\|\bar{\mathbf{z}}\|^2 / \text{tr}(\mathbf{W})$.
- (b) If Σ satisfies sphericity, then under H $F \sim F(m-1, (N-1)(m-1))$ exactly, for every $N \geq 2$.
- (c) Conversely, if the null distribution of F is $F(m-1, (N-1)(m-1))$ for every $N \geq 2$, then Σ satisfies sphericity.

Proof.

- (a) Since $\mathbf{C} = \mathbf{U}\mathbf{U}^\top$ and $\mathbf{U}$ has orthonormal columns, $\sum_j (\bar{y}_{\cdot j} - \bar{y})^2 = \|\mathbf{C}\bar{\mathbf{y}}\|^2 = \|\bar{\mathbf{z}}\|^2$, so $\text{SS}_T = N\|\bar{\mathbf{z}}\|^2 = \|\mathbf{q}\|^2$; likewise subject i 's contribution to SS_E is $\|\mathbf{z}_i - \bar{\mathbf{z}}\|^2$, summing to $\text{tr}(\mathbf{W})$. Under H the $\mathbf{z}_i$ are independent $N_{m-1}(\mathbf{0}, \Psi)$, so Lemma 14.2.3 gives the Wishart law of $\mathbf{W}$ and its independence of $\bar{\mathbf{z}}$. Finally $F = \{\|\mathbf{q}\|^2/(m-1)\} / \{\text{tr}(\mathbf{W})/(N-1)(m-1)\}$.
- (b) If $\Psi = \lambda \mathbf{I}_{m-1}$, then $\mathbf{q}/\sqrt{\lambda}$ is standard normal, so $\|\mathbf{q}\|^2/\lambda \sim \chi^2(m-1)$. And $\mathbf{W}/\lambda \sim W_{m-1}(N-1, \mathbf{I})$, which is the matrix of sums of squares and cross products of $(N-1)(m-1)$ independent standard normal variables, so $\text{tr}(\mathbf{W})/\lambda \sim \chi^2((N-1)(m-1))$. The two are independent, and λ cancels from the ratio, leaving the ratio of two independent chi-squared variables each divided by its degrees of freedom (Definition 4.1.2).
- (c) Write F_N for the statistic with N subjects. By (a),

$$F_N = \|\mathbf{q}\|^2 \cdot \frac{N-1}{\text{tr}(\mathbf{W})},$$

where $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$ has the *same* distribution for every N and is independent of $\mathbf{W}$. Since $\mathbf{W}/(N-1)$ is an average of $N-1$ independent $W_{m-1}(1, \Psi)$ matrices, the weak law of large numbers gives $\text{tr}(\mathbf{W})/(N-1) \rightarrow \text{tr} \Psi$ in probability, so by Slutsky's lemma $F_N \rightarrow \|\mathbf{q}\|^2 / \text{tr} \Psi$ in distribution. On the other hand $F(m-1, \nu) \rightarrow \chi^2(m-1)/(m-1)$ in distribution as $\nu \rightarrow \infty$, because its denominator chi-squared divided by ν tends to one in probability. If the exactness holds for every N , the two limits agree:

$$\|\mathbf{q}\|^2 \stackrel{d}{=} \frac{\text{tr} \Psi}{m-1} \chi^2(m-1).$$

Now match variances. By Theorem 4.2.3, $\text{Var}(\|\mathbf{q}\|^2) = 2 \text{tr}(\Psi^2)$, while the right-hand side has variance $2\{\text{tr} \Psi\}^2/(m-1)$. Equality says $(m-1) \text{tr}(\Psi^2) = \{\text{tr} \Psi\}^2$, that is $\epsilon(\Sigma) = 1$, and Lemma 33.3.1 makes that sphericity. ■

Idea. The repeated-measures F test does not require the occasions to be exchangeable, only that the $m-1$ orthonormalized within-subject contrasts be uncorrelated with equal variances — that the contrast space look spherical, which series of measurements in time almost never do.

33.3.3 Two corrections and one alternative

When sphericity fails, $\|\mathbf{q}\|^2$ and $\text{tr}(\mathbf{W})$ are still weighted sums of chi-squared variables, and matching their first two moments to scaled chi-squared variables gives a usable approximation (Box 1954).

Proposition 33.3.3. The moment-matched approximation

Under the assumptions of Theorem 33.3.2, with H true, write $\epsilon = \epsilon(\Sigma)$, $d_1 = m-1$ and $d_2 = (N-1)(m-1)$. Then

- (a) $E SS_T = \text{tr } \Psi$, $\text{Var } SS_T = 2 \text{tr}(\Psi^2)$, $E SS_E = (N - 1) \text{tr } \Psi$ and $\text{Var } SS_E = 2(N - 1) \text{tr}(\Psi^2)$.
- (b) The scaled chi-squared variable $g\chi^2(h)$ matching the first two moments of SS_T has $h = \epsilon d_1$ and $g = \text{tr}(\Psi^2)/\text{tr } \Psi$; the one matching SS_E has $h' = \epsilon d_2$ and the *same* g . Since SS_T and SS_E are independent by [Theorem 33.3.2\(a\)](#), replacing each by its matched scaled chi-squared gives the approximation $F \approx F(\epsilon d_1, \epsilon d_2)$, which is exact when $\epsilon = 1$.

Proof.

- (a) The first two are Theorem 4.2.3 with $\mathbf{A} = \mathbf{I}$ and mean zero. Writing $\mathbf{W} = \sum_{l=1}^{N-1} \mathbf{w}_l \mathbf{w}_l^\top$ with $\mathbf{w}_l$ independent $N_{m-1}(\mathbf{0}, \Psi)$ (Lemma 14.2.3), $\text{tr } \mathbf{W} = \sum_l \|\mathbf{w}_l\|^2$ is a sum of $N - 1$ independent copies of $\|\mathbf{q}\|^2$, so its mean and variance are $N - 1$ times theirs.
- (b) Matching $gh = \text{tr } \Psi$ and $2g^2h = 2 \text{tr}(\Psi^2)$ gives $g = \text{tr}(\Psi^2)/\text{tr } \Psi$ and $h = \{\text{tr } \Psi\}^2/\text{tr}(\Psi^2) = \epsilon d_1$. For SS_E both moments are multiplied by $N - 1$, so g is unchanged and $h' = (N - 1)h = \epsilon d_2$. Since $d_2/d_1 = h'/h$, the ratio $(SS_T/d_1)/(SS_E/d_2)$ becomes $\{\chi^2(h)/h\}/\{\chi^2(h')/h'\}$, an $F(h, h')$ variable, the two chi-squared variables being independent by [Theorem 33.3.2\(a\)](#); at $\epsilon = 1$ this is the exact result of [Theorem 33.3.2\(b\)](#). ■

By [Lemma 33.3.1](#), $\epsilon \geq 1/(m - 1)$, so the approximating distribution is never more dispersed than $F(d_1/(m - 1), d_2/(m - 1)) = F(1, N - 1)$, the largest that [Proposition 33.3.3\(b\)](#) can ask for. Referring F to $F(1, N - 1)$ is therefore conservative within Box's approximation, and Geisser and Greenhouse (1958) proved it exactly conservative, for every Σ and N . This is the **lower-bound** correction, turned into a recommendation by Greenhouse and Geisser (1959).

Part (b) becomes a procedure once ϵ is estimated from the sample covariance matrix $\mathbf{S}$ of the $\mathbf{y}_i$.

- The **Greenhouse–Geisser** correction refers F to $F(\hat{\epsilon}d_1, \hat{\epsilon}d_2)$ with the plug-in $\hat{\epsilon} = \epsilon(\mathbf{S})$, which is biased downward, so the test is conservative.
- The **Huynh–Feldt** correction removes the leading bias. For a single group of N subjects it uses

$$\tilde{\epsilon} = \min\left\{1, \frac{N(m - 1)\hat{\epsilon} - 2}{(m - 1)\{(N - 1) - (m - 1)\hat{\epsilon}\}}\right\}, \quad (33.3.3)$$

which exceeds $\hat{\epsilon}$ and gives a less conservative test; its own bias is upward when ϵ is small.

- The **lower-bound** correction takes $\epsilon = 1/(m - 1)$: safe, and usually too conservative to be useful.

A third option avoids the issue. Since the $\mathbf{z}_i$ are independent $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \Psi)$ with Ψ unrestricted, H says a multivariate normal mean is zero, and the classical test is Hotelling's.

Proposition 33.3.4. The multivariate test

Under the assumptions of [Theorem 33.3.2](#) with $N \geq m$, let $\mathbf{S}_z = \mathbf{W}/(N - 1)$ and

$$T^2 = N \bar{\mathbf{z}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}}.$$

Then T^2 does not depend on the choice of $\mathbf{U}$, nor on replacing $\mathbf{U}$ by $\mathbf{UK}$ for any nonsingular $(m-1) \times (m-1)$ matrix $\mathbf{K}$; and under H ,

$$\frac{N-m+1}{(N-1)(m-1)} T^2 \sim F(m-1, N-m+1),$$

whatever Σ is. The distributional statement is Hotelling's and is not proved here; see Anderson (2003, §5.2).

Proof. Only the invariance is proved. Replacing $\mathbf{U}$ by $\mathbf{UK}$ replaces $\bar{\mathbf{z}}$ by $\mathbf{K}^\top \bar{\mathbf{z}}$ and $\mathbf{S}_z$ by $\mathbf{K}^\top \mathbf{S}_z \mathbf{K}$, leaving T^2 unchanged; two orthonormal bases differ by an orthogonal $\mathbf{K}$. The requirement $N \geq m$ makes $\mathbf{S}_z$, a Wishart matrix on $N-1 \geq m-1$ degrees of freedom, invertible with probability one. ■

Exactness under any Σ sounds decisive, but it costs power: the multivariate test estimates a whole covariance matrix and has $N-m+1$ denominator degrees of freedom rather than $(N-1)(m-1)$. When sphericity nearly holds the univariate test is much more powerful, when it fails badly the two are comparable, and when $N < m$ the multivariate test does not exist.

33.3.4 An example, four ways

Example. Grip strength over six visits

Twenty-four patients have grip strength measured, in kilograms, at six weekly visits, simulated from $N_6(\boldsymbol{\mu}, \Sigma)$ with Σ first-order autoregressive, $\sigma_{jk} = 4 \times 0.6^{|j-k|}$, and a mean drifting slowly upward; the observed occasion means are 47.818, 48.099, 49.056, 49.177, 49.561 and 49.263 kg. The true Box measure is $\epsilon = 0.700$: adjacent visits correlate 0.6 and visits five weeks apart only 0.6^5 , so adjacent differences are much less variable than distant ones. The four analyses of H give

Analysis	statistic	degrees of freedom	p
univariate, uncorrected	4.770	5, 115	0.0005
Greenhouse–Geisser, $\hat{\epsilon} = 0.645$	4.770	3.23, 74.2	0.0035
Huynh–Feldt, $\tilde{\epsilon} = 0.763$	4.770	3.81, 87.7	0.0019
multivariate	3.605	5, 19	0.0184

All four reject, but the evidence differs by a factor of nearly forty between first and last. Distrust the uncorrected test: its nominal degrees of freedom are too large by the factor ϵ .

```
import numpy as np
from scipy import stats
```

```
N, m = 24, 6
```

```

mu_true = np.array([48.0, 48.5, 48.9, 49.2, 49.4, 49.5]) # kg, a slow improvement
sigma, rho_true = 2.0, 0.6
lags = np.abs(np.subtract.outer(np.arange(m), np.arange(m)))
Sigma_true = sigma**2 * rho_true**lags # AR(1) within a patient
rng = np.random.default_rng(3303)
Yobs = np.round(mu_true + rng.multivariate_normal(np.zeros(m), Sigma_true, N), 2)

def contrast_basis(m):
    """Orthonormal basis U of the space of contrasts among m repeated measures."""
    Q, _ = np.linalg.qr(np.column_stack([np.ones(m), np.eye(m)[:, 1:]])
    return Q[:, 1:] # columns orthonormal and
                    orthogonal to 1

def epsilon(Psi):
    """Box's measure of departure from sphericity, 1/(m-1) <= eps <= 1."""
    k = Psi.shape[0]
    return np.trace(Psi)**2 / (k * np.sum(Psi * Psi))

U = contrast_basis(m)
Z = Yobs @ U # N x (m-1) contrasts, one row
              per patient
zbar = Z.mean(axis=0)
Sz = np.cov(Z, rowvar=False, ddof=1) # Wishart, (N-1) degrees of
                                     freedom

F_uncorrected = N * (zbar @ zbar) / np.trace(Sz)
eps_hat = epsilon(Sz)
eps_hf = min(1.0, (N * (m - 1) * eps_hat - 2)
             / ((m - 1) * ((N - 1) - (m - 1) * eps_hat)))
d1, d2 = m - 1, (N - 1) * (m - 1)
p_un = stats.f.sf(F_uncorrected, d1, d2)
p_gg = stats.f.sf(F_uncorrected, d1 * eps_hat, d2 * eps_hat)
p_hf = stats.f.sf(F_uncorrected, d1 * eps_hf, d2 * eps_hf)
T2 = N * zbar @ np.linalg.solve(Sz, zbar) # Hotelling's statistic
F_mv = (N - m + 1) / ((N - 1) * (m - 1)) * T2
p_mv = stats.f.sf(F_mv, m - 1, N - m + 1)
print(f"F = {F_uncorrected:.3f} on ({d1}, {d2}) df, p = {p_un:.4f}")
print(f"Greenhouse-Geisser eps = {eps_hat:.3f}, p = {p_gg:.4f}")
print(f"Huynh-Feldt eps = {eps_hf:.3f}, p = {p_hf:.4f}")
print(f"multivariate F({m - 1}, {N - m + 1}) = {F_mv:.3f}, p = {p_mv:.4f}")

```

[Run this code online](#)

How much does the failure of sphericity cost? [Figure 33.3.1](#) answers with a simulation under H . Panel (a) shows the null distribution of the uncorrected F : more dispersed than $F(5, 115)$, with a heavier right tail, and tracking [Proposition 33.3.3](#) closely. Panel (b) traces the level of a nominal 5% test as the autoregressive parameter grows. At $\rho = 0$ it is 0.050, as [Theorem 33.3.2\(b\)](#) requires; by $\rho = 0.8$, where $\epsilon = 0.562$, it is 0.085. The Greenhouse–Geisser test is conservative throughout (0.038 to 0.049), the Huynh–Feldt test close to nominal (0.049 to 0.056), and the multivariate test exact at 0.049 for every ρ . A spherical covariance of the form [Lemma 33.3.1\(iv\)](#) with unequal a_j , far from compound symmetric, gives the uncorrected test a level of 0.050: sphericity, not compound symmetry, is what matters.

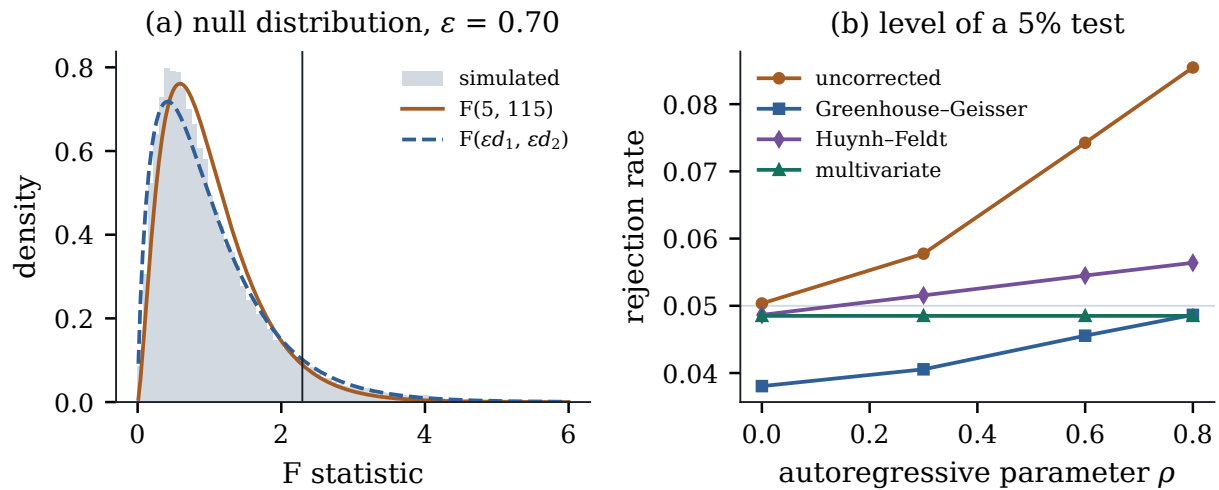


Figure 33.3.1: (a) The null distribution of the uncorrected F statistic with autoregressive errors, against the nominal $F(5, 115)$ density and the moment-matched $F(\epsilon d_1, \epsilon d_2)$ of Proposition 33.3.3, with the nominal 5% point marked. (b) Simulated level of four tests as ρ grows.

33.3.5 The modern treatment

The corrections above patch a fixed analysis; the mixed-model approach fits the covariance instead. Write the data as a linear mixed model (Definition 32.2.1) with a fixed effect for each occasion and a within-subject covariance $\mathbf{R}$ of chosen shape, estimate its parameters by restricted maximum likelihood (Definition 32.4.1), and test the fixed effects by a Wald statistic with estimated degrees of freedom (Proposition 32.5.1). Compound symmetric $\mathbf{R}$ reproduces the uncorrected univariate F (Exercise (B2)), unstructured $\mathbf{R}$ the multivariate test up to the degrees-of-freedom approximation, and anything between is a model the classical scheme has no room for. Unbalanced data and missing visits are handled too, whereas Eq. 33.3.1 and T^2 need a complete rectangle. The price is that the covariance must be chosen, and the choice affects the standard errors: Section 33.4.

Remark (Between-subjects factors). With a between-subjects factor the within-subject analysis is the same, applied to the pooled within-group covariance, and the univariate tests are exact iff the pooled Ψ is spherical *and* the groups share a common Σ . Mauchly's (1940) test of sphericity itself, which most software reports, is sensitive to nonnormality and, like any preliminary test, distorts the level of what follows.

33.3.6 Exercises

33.3.6.1 A. Check your understanding

Exercise. A1

Show that every 2×2 covariance matrix satisfies sphericity, so that with $m = 2$ the repeated-measures F test is exact whatever Σ is. Identify the test in that case.

Solution. With $m = 2$, $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$ is 1×1 , hence a multiple of $\mathbf{I}_1$. By Theorem 33.3.2(b) the test is exact. It is the paired t test: the single contrast is $(y_{i1} - y_{i2})/\sqrt{2}$, and F is the

square of the paired t statistic on $N - 1$ degrees of freedom.

Exercise. A2

For $m = 3$ and an autoregressive covariance $\sigma_{jk} = \rho^{|j-k|}$ with $0 < \rho < 1$, compute Ψ for the basis $\mathbf{u}_1 = (1, -1, 0)^\top / \sqrt{2}$, $\mathbf{u}_2 = (1, 1, -2)^\top / \sqrt{6}$ and show that sphericity fails. Which pairwise difference has the largest variance?

33.3.6.2 B. Practice

Exercise. B1

Show directly that compound symmetry $\Sigma = \sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$ gives $\Psi = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$, and hence that the split-plot analysis of Theorem 33.1.2 is the repeated-measures F test with $\epsilon = 1$. Which λ of Lemma 33.3.1 does it correspond to?

Solution. $\mathbf{U}^\top \mathbf{1} = \mathbf{0}$, so $\Psi = \mathbf{U}^\top \Sigma \mathbf{U} = \sigma^2(1 - \rho)\mathbf{U}^\top \mathbf{U} = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$. Hence $\epsilon = 1$ and $\lambda = \sigma^2(1 - \rho)$, which is the subplot stratum eigenvalue λ_s of Lemma 33.1.1. In Lemma 33.3.1(iv) take $\mathbf{a} = (\sigma^2\rho/2)\mathbf{1}$.

Exercise. B2

Consider the mixed model $y_{ij} = \mu_j + u_i + e_{ij}$ with u_i and e_{ij} independent normal, of variances σ_u^2 and σ_e^2 . Show that restricted maximum likelihood gives $\hat{\sigma}_e^2 = \text{MS}_E$ and $\hat{\sigma}_u^2 = (\text{MS}_S - \text{MS}_E)/m$ when this is nonnegative, where MS_S is the between-subject mean square, and that the Wald statistic for $H : \mu_1 = \cdots = \mu_m$, divided by $m - 1$, is the uncorrected F of Eq. 33.3.1.

33.3.6.3 C. Going deeper

Exercise. C1

For $m = 4$, $N = 20$ and Σ compound symmetric, compare by simulation the power of the uncorrected univariate test, the Greenhouse–Geisser test and the multivariate test of Proposition 33.3.4 against a linear trend in the occasion means. Repeat with an autoregressive Σ of the same total variance. Which would you choose if you knew the alternative were a linear trend, and why is a single contrast better still?